L1: Introduction to Machine Learning

Core Concepts, Applications & Foundations | Instructor: Arun Rajkumar

> **Key Takeaway:** Machine learning is fundamentally the practice of using **data** and **mathematical models** to enable algorithms to **generalise** to unseen inputs. It differs strictly from procedural programming (fixed rules) and memorisation (exact lookup).

1. Applications & Industry Trends

Machine learning, deep learning, and artificial intelligence have experienced explosive growth since 2016 (evident in global search trends). These technologies drive key digital transformation trends highlighted by Forbes, including conversational AI, predictive analytics, and autonomous systems.

Human Sensory Emulation

- **Computer Vision:** Giving machines visual perception to process images and perform object recognition (e.g., distinguishing trees or faces in photo albums).
- **Speech & Natural Language:** Processing unstructured spoken input across diverse accents and speakers to understand natural language.
- **Text & Semantics:** Extracting semantic meaning from millions of internet documents and recognizing handwritten text.
- **Computational Brain Modeling:** The long-term goal of modeling the brain's complex neural networks to understand learning from an abstract computational perspective.

Real-World Use Cases

- **E-Commerce & OTT Streaming:** Recommendation systems (Amazon, Flipkart, Netflix) that predict user preferences based on similar user profiles.
- **Finance & Weather:** Stock market trend forecasting and precipitation/rainfall prediction using historical environmental parameters.
- **Everyday Utilities:** Automated email spam filters, social network connection suggestions (LinkedIn link prediction), face clustering in photo albums, and voice/instrument separation in audio files.

2. What Machine Learning Is vs. What It Is Not

To clearly define machine learning, it is helpful to contrast it with conventional computational approaches:

Approach	Mechanism	Limitation / Goal
Procedural Programming	Applies fixed, step-by-step rules to transform inputs to outputs (e.g., tax calculation from salary).	No learning occurs. Restricted strictly to explicitly programmed logic.
Memorisation	Stores known examples directly (e.g., memorising 2,000 tree images or assignment solutions).	Fails completely when presented with new, unseen inputs.
Machine Learning	Extracts underlying patterns and structures from **Data** using mathematical models.	Achieves **Generalisation**—performing accurately on previously unseen data.

3. Core Mathematical Foundations

Machine learning is not magic or intuition—it is built entirely on mathematical foundations that translate raw observations into reliable computational decisions:

- **Linear Algebra:** Provides the framework for representing data, vectors, matrices, and linear geometric structures (e.g., fitting lines or planes to data).
- **Probability & Statistics:** Offers the mathematical language to model real-world uncertainty, noise, sensor measurement errors, and incomplete information.
- **Calculus & Optimization:** Enables defining loss/cost metrics to systematically find optimal decisions and model parameters that minimize error.